Model Diagnostics: Theoretical Foundations

Md. Ikramul Haq Khan

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Standalone companion to the complete regression diagnostics reference. Personal study notes prepared with AI assistance; not an official course handout or organizational SOP.

1 Start with the scientific question

Diagnostics assess whether limitations of a fitted model threaten a specific conclusion. Before fitting, record the study design, outcome, observational unit, target population and estimand. Description estimates patterns in observed data; prediction targets future outcomes; causal inference targets effects of interventions and requires identification assumptions beyond residual checks. For example, predicting industrial stack loss from operating conditions is different from claiming that changing one condition causes an outcome change. A clean residual plot cannot establish absence of confounding.

2 Residual, error, leverage and influence

Let X have full column rank p, including the intercept when present, with $n > p$. Under the classical conditional mean and covariance model,

$$Y = X\beta + \varepsilon, \quad E(\varepsilon \mid X) = 0, \quad \operatorname{Var}(\varepsilon \mid X) = \sigma^2 I_n.$$

$$\widehat{\beta} = (X^T X)^{-1} X^T Y, \quad H = X(X^T X)^{-1} X^T, \quad \widehat{Y} = HY, \quad e = (I - H)Y.$$

2.1 Residual versus unobserved error

The error $\varepsilon_i = Y_i - E(Y_i \mid X)$ is relative to the true conditional mean; the observed residual $e_i = y_i - \widehat{y}_i$ is relative to the estimated mean. In a scatterplot it is a vertical discrepancy. If a fitted score is 70 and the observed score is 85, the residual is 15 points. Because $(I - H)X = 0$, $e = (I - H)\varepsilon$. Symmetry and idempotence give

$$E(e \mid X) = 0, \quad \operatorname{Var}(e \mid X) = \sigma^2(I - H),$$

$$\operatorname{Var}(e_i \mid X) = \sigma^2(1 - h_{ii}), \quad \operatorname{Cov}(e_i, e_j \mid X) = -\sigma^2 h_{ij}.$$

Thus residuals are generally correlated and have unequal variances even when errors are uncorrelated with equal variance. Also $X^T e = 0$; with an intercept $\sum_i e_i = 0$. Zero average residual is an algebraic consequence, not evidence that the model is correct.

2.2 Leverage: unusual predictors

Leverage h_{ii} depends only on the design. Since $\operatorname{tr}(H) = p$, average leverage is p/n. In simple regression with an intercept,

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}.$$

In multiple regression an unusual combination of predictors can have high leverage even when no individual predictor is extreme. The heuristic $h_{ii} > 2p/n$ or $3p/n$ flags cases for review; it is not a hypothesis test.

2.3 Studentized residuals and outliers

Put $s^2 = \sum e_i^2/(n-p)$. The internally studentized residual is

$$r_i = \frac{e_i}{s\sqrt{1-h_{ii}}}.$$

It is not exactly t_{n-p} distributed. If $s_{(i)}$ estimates error SD after deleting case i, the externally studentized residual is

$$t_i = \frac{e_i}{s_{(i)}\sqrt{1-h_{ii}}}.$$

Under the normal linear model, for a prespecified observation, $t_i \sim t_{n-p-1}$, assuming the deleted design remains full rank and $n > p+1$. For searching all n observations, use multiplicity-aware assessment, for example Bonferroni cutoff $t_{1-\alpha/(2n),n-p-1}$. A regression outlier has an unusual outcome conditional on its predictors; an extreme marginal outcome need not be an outlier.

2.4 Influence: sensitivity to deletion

An influential observation substantially changes fitted quantities when omitted. With $\widehat{y}_{j(i)}$ denoting predictions from the deleted fit,

$$D_i = \frac{\sum_j (\widehat{y}_j - \widehat{y}_{j(i)})^2}{ps^2} = \frac{r_i^2}{p}\frac{h_{ii}}{1-h_{ii}}.$$

Cook's distance combines residual discrepancy and leverage. Rules such as $D_i > 4/n$ are screening conventions, not universal significance thresholds. DFBETAS measure standardized changes in individual coefficients; DFFITS measures a standardized change in the case's fitted value. The deleted prediction residual is

$$d_i = y_i - \widehat{y}_{i(i)} = \frac{e_i}{1-h_{ii}}.$$

High leverage can pull the fit toward a point, concealing disagreement in the raw residual. Deletion diagnostics reveal this sensitivity. Multiple unusual observations can mask each other; single-case deletion is not sufficient in every dataset.

3 Normal Q–Q plots: quantile, percentile, quartile

A quantile is a distribution cutoff: $Q(p) = \inf\{y : F(y) \geq p\}$. The 90th percentile is $Q(0.90)$. Quartiles are $Q(0.25)$, $Q(0.50)$ and $Q(0.75)$. Q–Q means quantile–quantile, not quartile–quartile.

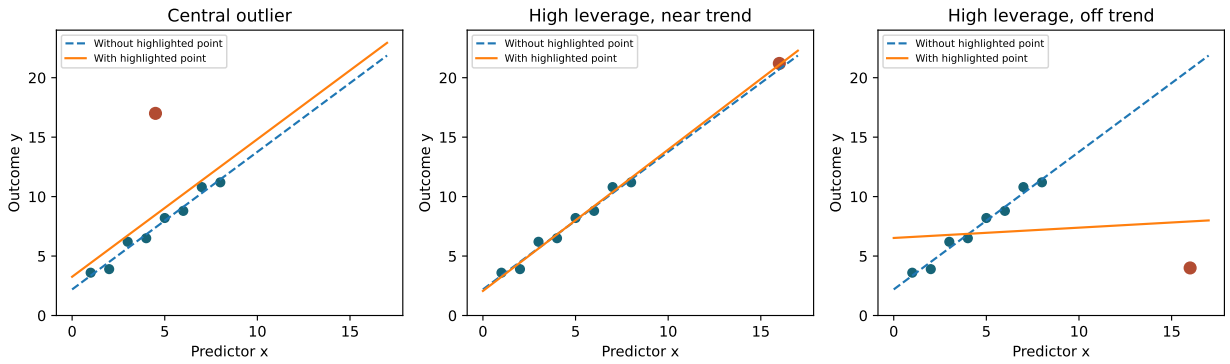


Figure 1: Computed OLS fits for three illustrative configurations. The dashed line excludes the highlighted observation. Position in predictor space and change in the fitted line answer different questions.

For a normal variable, $Q(p) = \mu + \sigma\Phi^{-1}(p)$. This linear relationship motivates comparing ordered observations $e_{(i)}$ with $z_i = \Phi^{-1}((i - 0.5)/n)$; software plotting positions may differ slightly. The horizontal axis is typically theoretical normal quantiles and the vertical axis observed ordered residuals. A straight reference line need not be $y = x$. Approximate linearity supports normal-like shape; systematic asymmetry suggests skewness. Relative to a suitable central line, heavy tails typically place the lower tail below and upper tail above it; light tails show the reverse. Finite-sample fluctuations, leverage and parameter estimation complicate interpretation. The plot does not establish independence, constant variance, correct mean structure or causality.

4 Why diagnostics require judgment

The reference mathematics is precise; adequacy is purpose-dependent. Finite data cannot prove a model true. A formal test fixes an error-rate convention for specified alternatives, but small samples can miss important departures and large samples can detect negligible ones. Repeated testing and adapting the model to diagnostics can also distort subsequent inference. Distinguish three statements: (1) a departure is detectable; (2) it changes the estimand's estimate or uncertainty; (3) it changes a scientific decision. Quantify the latter two through sensitivity analyses and validation. Report what was checked and what changed, rather than declaring that all assumptions were verified.

5 Mathematical tests and their limits

5.1 Specified additional mean terms: partial F test

Compare nested reduced and full models fitted to identical observations, with q additional parameters and p_F full-model parameters:

$$H_0 : \beta_{\text{added}} = 0, \qquad F = \frac{(\text{SSE}_R - \text{SSE}_F)/q}{\text{SSE}_F/(n - p_F)}.$$

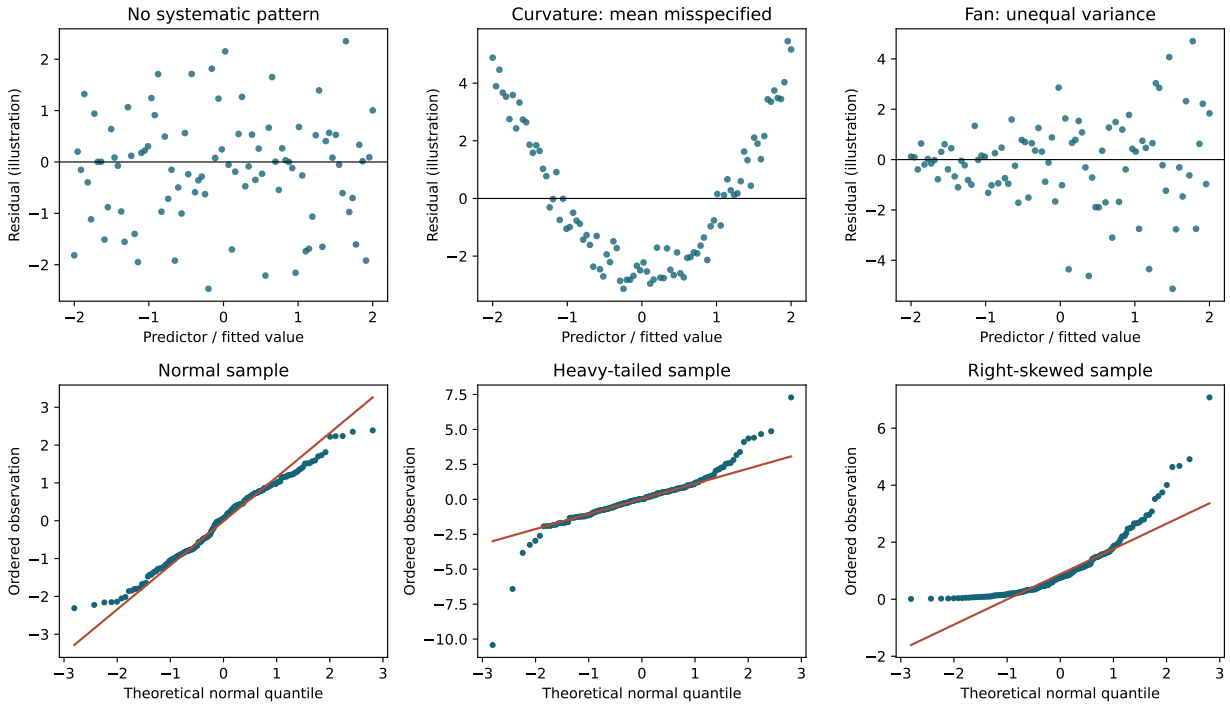


Figure 2: Simulated patterns, not empirical findings. Top: residual signatures to investigate. Bottom: Q–Q comparisons using independent simulated samples to isolate distribution shape; fitted regression residuals have additional dependence.

Under a correct full mean model with independent normal, constant-variance errors, $F \sim F_{q,n-p_F}$ under H_0. For example, compare $y \sim x$ against $y \sim x + x^2$ to test a specified curvature term. Rejection supports the added terms; nonrejection does not prove linearity. RESET adds powers of fitted values as an omnibus specification screen and cannot identify the correct replacement model. Use robust Wald inference when justified if the classical covariance model fails.

5.2 Breusch–Pagan / Koenker variance tests

The null is constant conditional error variance. An auxiliary regression relates squared residuals to chosen variance predictors. In the usual studentized Koenker formulation, $LM = nR^2$ from this auxiliary regression is asymptotically χ^2_q under regularity and suitable moments; q excludes its intercept. The original BP version uses a different normalization and stronger distributional assumptions. In R, `lmtest::bptest(fit)` defaults to the studentized version. Neither version detects every variance pattern. Mean misspecification can also produce apparent variance patterns.

5.3 Serial dependence: Breusch–Godfrey

For genuinely ordered data, regress residuals on original regressors and q lagged residuals. Under regularity and a valid mean/dynamic specification, an LM statistic based on the effective auxiliary sample is asymptotically χ^2_q for the null of no serial correlation through those lags. Software differs in treatment of initial lags. Use `bgtest(fit, order=q)` only when ordering is meaningful. Serial tests are not substitutes for cluster-aware analysis.

5.4 Normality and simulation calibration

Shapiro–Wilk tests normality for an independent normal sample. Applying its usual reference distribution directly to fitted residuals ignores their design-dependent joint distribution. Treat an ordinary residual-based p-value cautiously. A parametric simulation can calibrate a residual statistic under the fitted normal linear model: simulate normal errors at fixed X, refit, recompute the same statistic, and compare with the observed value. This explicitly includes fitting and leverage in the reference distribution. It tests a specified normal-model reference, not all possible model errors.

5.5 Assumptions needed for different conclusions

OLS computation needs full column rank, not normality. Conditional unbiasedness needs the correct linear conditional mean and zero conditional error mean. Gauss–Markov efficiency additionally uses spherical covariance. Classical exact small-sample normal-theory inference uses normal errors. Suitable robust covariance estimators can address some covariance violations asymptotically; they do not repair confounding, a wrong mean, or unreliable data.

6 A practical diagnostic workflow

1. Define the target and design. Identify repeated measures, sampling weights, clusters, missingness and temporal order before choosing a model.

2. Check units, coding, impossible values and measurement processes. Preserve legitimate unusual observations.

3. Examine outcome/predictor distributions and scientifically plausible nonlinearities or interactions.

4. Fit a defensible initial model. Inspect residuals versus fitted values, each predictor and relevant time/group variables; inspect a Q–Q plot.

5. Quantify leverage, externally studentized residuals and influence. Investigate flagged cases and compare estimates with and without them as sensitivity analysis, not automatic exclusion.

6. Use targeted tests with stated hypotheses and assumptions. Avoid testing until all p-values exceed 0.05.

7. Compare scientifically justified alternatives. For prediction, use held-out or cross-validated evaluation with splits matching future deployment; keep tuning separate from final evaluation.

8. Report estimates, uncertainty, departures, sensitivity and remaining limitations.

For clustered or survey data, ordinary H, Cook's distance and iid reference tests do not directly provide design-valid inference. Use the appropriate weighted/design model and cluster- or PSU-level sensitivity checks. For missing data, complete-case residuals describe the analyzed subset; they do not establish ignorability of missingness. For counts, binary outcomes or infection trajectories, choose appropriate likelihoods and dependence/observation models rather than forcing Gaussian diagnostics onto every outcome.

7 Practice with feedback

1. A point lies far to the right but on the trend. Is it necessarily influential? *Answer:* It has high leverage; actual influence requires sensitivity of fitted quantities. Check deletion.

2. A Q–Q plot is straight but the residual plot fans outward. Which assumption is questionable? *Answer:* Constant conditional variance. A Q–Q plot alone cannot assess it.

3. If residuals sum to zero, is the conditional mean correctly specified? *Answer:* No; this holds algebraically for OLS with an intercept.

4. A BP test gives $p = 0.30$. Can you claim equal variances? *Answer:* No; report insufficient evidence against the tested null and assess precision, power and graphics.

5. A coefficient changes sign after deleting one valid case. What next? *Answer:* Report sensitivity, inspect coding and design, examine specification and data coverage; do not delete solely to obtain a desired conclusion.

6. Lab task: list the three largest Cook's distances, compare the full and deleted coefficient estimates, and explain whether the substantive conclusion changes. *Feedback:* Rank cases from computed diagnostics; do not invent an influential case from a remembered textbook result. Compare changes on meaningful scales, with uncertainty.

8 Expert resources and a reading pathway

This is a curated, broad pathway, not an exhaustive bibliography. Links were checked through source discovery on October 29, 2026. Package behavior can change; inspect installed help and session information.

1. **Penn State STAT 501**: Lesson 11: Influential Points; Lesson 4: Model Assumptions. Start here for worked distinctions among outliers, leverage and influence.
2. **NIST/SEMATECH**: Regression diagnostics reference. Use for a broad catalogue of diagnostic ideas and applied checks.
3. **R Core documentation**: plot.lm; influence.measures. Read for exact definitions used by software.
4. **lmtest authors' documentation**: bptest; bgtest; resettest. Study hypotheses, defaults and reference distributions before using tests.
5. **Frank E. Harrell Jr.**: Regression Modeling Strategies course notes. Continue here for specification, nonlinear effects, missing data, overfitting and validation, especially in biomedical work.
6. **R datasets documentation**: Brownlee's Stack Loss Plant Data. Read the primary software dataset record before interpreting the lab.

Suggested sequence: definitions and figures; Penn State examples; execute the lab; consult R/lmtest references; then Harrell's broader modeling strategy. A defensible report connects the scientific target, evidence of departures, and sensitivity of conclusions.